

THE DUAL NATURE OF MATTER & RADIATION: A VISUAL REVISION GUIDE

The Core Concept: Light's Dual Nature

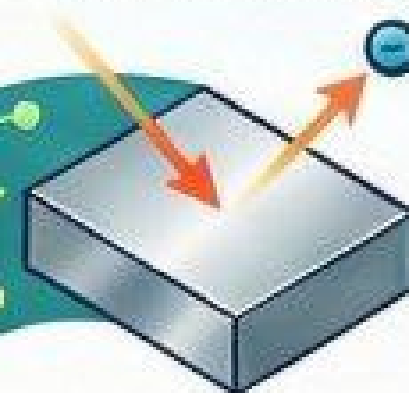


Wave Nature
Interference and diffraction demonstrate light's wave nature.

Light behaves as a wave in some phenomena and as a particle in others. This is called the Dual Nature of Radiation.

Particle Nature
The Photoelectric Effect demonstrates its particle nature.

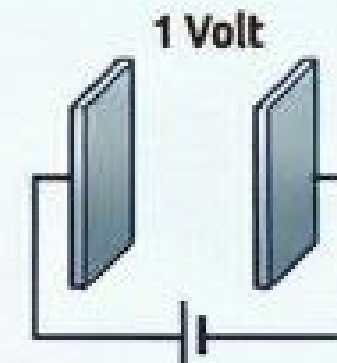
Understanding the Basics



Work Function (Φ_0 or W)
The minimum amount of energy required to eject a free electron from a metal surface.

Work Functions of Different Metals

- Cesium: 2.14 eV
- Potassium: 2.3 eV
- Platinum: 5.65 eV



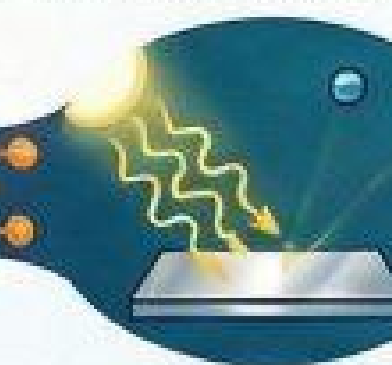
What is an Electron Volt (eV)?
A small unit of energy, it's the energy an electron gains when accelerated by a potential difference of 1 Volt.

Energy Conversion
 $1 \text{ eV} = 1.6 \times 10^{-19} \text{ Joules}$

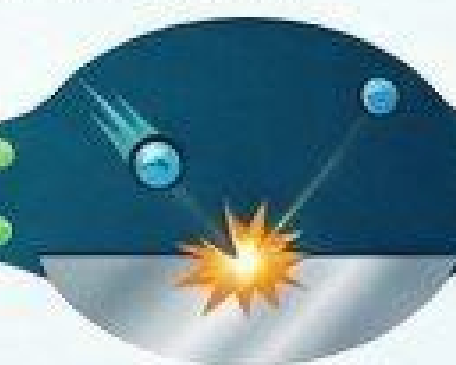
Four Ways to Eject an Electron (Electron Emission)



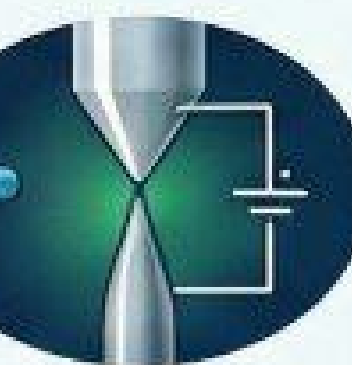
Thermionic Emission
Ejecting electrons by heating a metal (providing thermal energy).



Photoelectric Emission
Ejecting electrons (photoelectrons) by shining light of a suitable frequency on a metal.



Secondary Emission
A fast-moving "primary" electron strikes a surface, knocking out a "secondary" electron.



Field Emission
Pulling an electron out of a metal surface using a very strong electric field.

Einstein's Explanation: The Photon

Light is made of Photons
Einstein proposed that light consists of discrete energy packets called photons.

Photon Energy

$$E = hf$$

h = Planck's Constant ($6.62 \times 10^{-34} \text{ J}\cdot\text{s}$)

Properties of a Photon

Photons are chargeless, have zero rest mass, and travel at the speed of light.

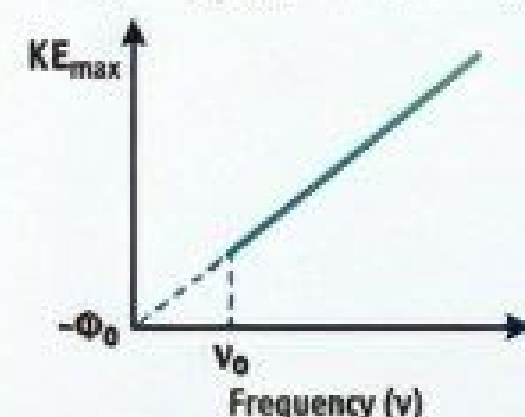
Einstein's Photoelectric Equation

$$hf = \Phi_0 + KE_{\text{max}}$$

Key Graphs of the Photoelectric Effect

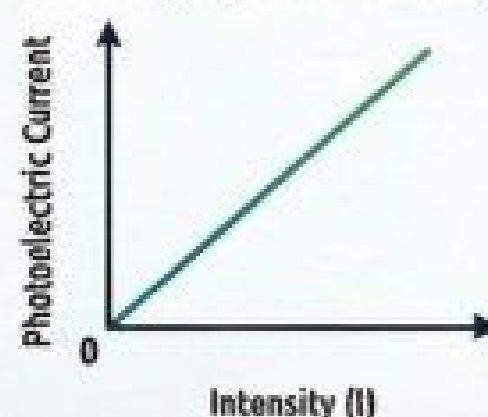
Kinetic Energy vs. Frequency

Straight line, starts at threshold frequency (ν_0), y-intercept is $-\Phi_0$.



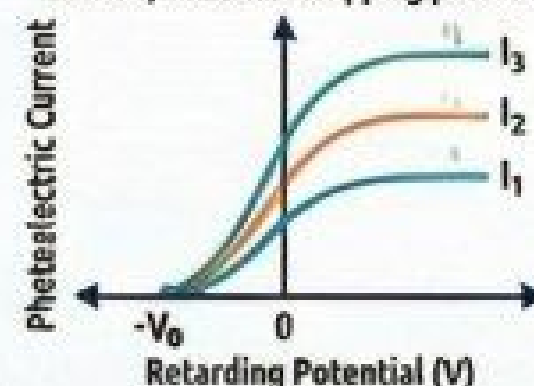
Current vs. Intensity

Straight line passing through origin; current is proportional to intensity.



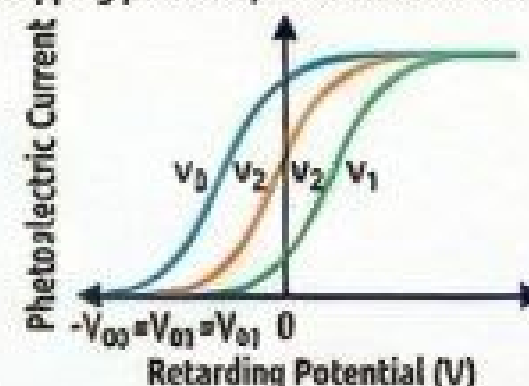
Effect of Intensity on Stopping Potential

Higher intensity = higher saturation current, but same stopping potential.



Effect of Frequency on Stopping Potential

Higher frequency = more negative stopping potential, same saturation current.



De Broglie's Hypothesis: Matter has Waves Too

Every Moving Particle has a Wave
De Broglie proposed that any moving matter has an associated matter wave.



Why Don't We See Waves in Daily Life?

For large mass (m), wavelength (λ) is incredibly small and undetectable.

De Broglie Wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Wavelength of an Accelerated Electron

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ Angströms (\AA)}$$

For an electron accelerated by a potential difference V .